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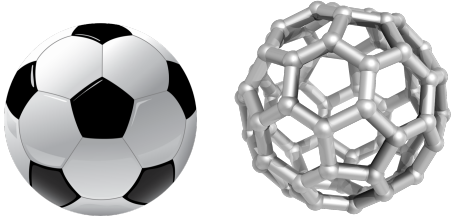
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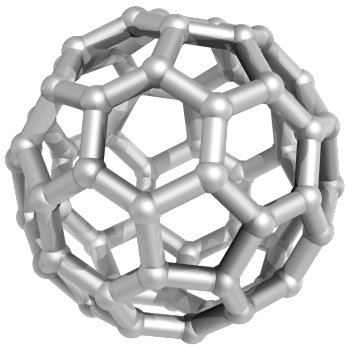
How to make a football bipartite

Matěj Stehlík

Rio COMB — First edition
UERJ, Rio de Janeiro, 26/10/2026



Fullerene molecules



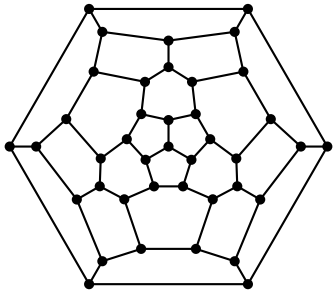
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- Each carbon atom has bonds to exactly three other carbon atoms.
- The carbon atoms form rings of five or six atoms.
- Existence predicted by Osawa in 1970.
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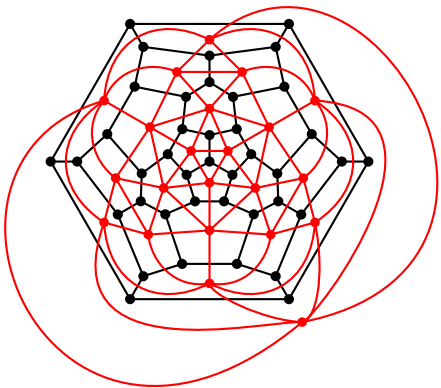
Fullerene graphs and their duals



A fullerene graph is:

- simple, 3-connected
- plane
- cubic
- all faces have size 5 or 6.

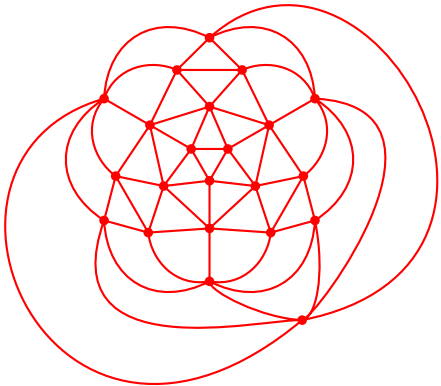
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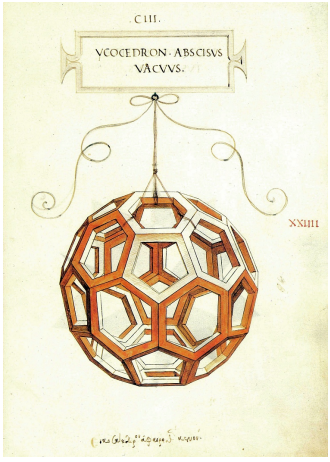
Its dual is:

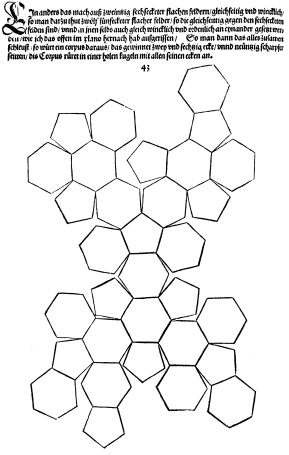
- simple
- plane
- triangulation
- all vertices have degree 5 or 6.

Named after Buckminster Fuller, pioneer of geodesic domes. . .



. . . fullerenes were known much earlier





Why are fullerene graphs (and their triangulations) interesting?

- They are a very natural class: having degree at most 6 is a combinatorial analogue of non-negative Gaussian curvature.

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- They are a very natural class: having degree at most 6 is a combinatorial analogue of non-negative Gaussian curvature.
- Their structure may shed some light on the chemical properties of the corresponding fullerene molecule
- By forbidding large faces, we can (sometimes) get much stronger results than in the general case.

The shape of a fullerene graph

- In a convex polyhedral metric, the neighbourhood of each point is isometric to a cone (possibly a plane)

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- Similarly for triangulation with all vertices of degree at most 6.

The four-colour theorem

Theorem (Appel and Haken 1977)

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- Tait showed that a plane triangulation is 4-colourable iff the dual graph is 3-edge colourable.
- He then convinced himself (and everyone else) that every 3-connected cubic planar graph has a Hamilton cycle, which leads to a 3-edge-colouring.

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- Kardoš's proof is computer-assisted.

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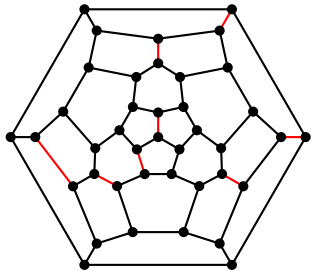
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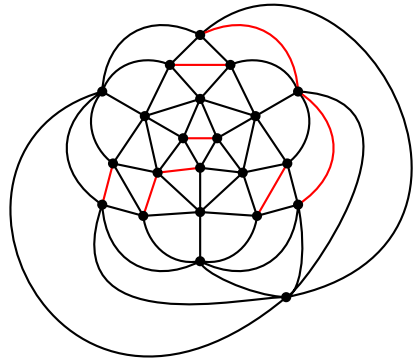
Theorem (Faria, Klein and MS 2012)

If G is a fullerene graph on n vertices, then $\tau_{\text{odd}}(G) \leq \sqrt{12n/5}$, with equality iff $n = 60k^2$ and G has the full icosahedral automorphism group.

Odd cycle transversals and their duals

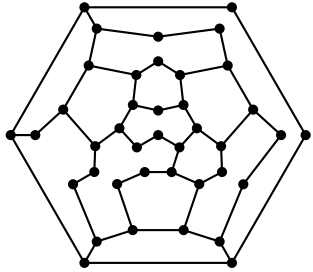


$G - J$ is bipartite

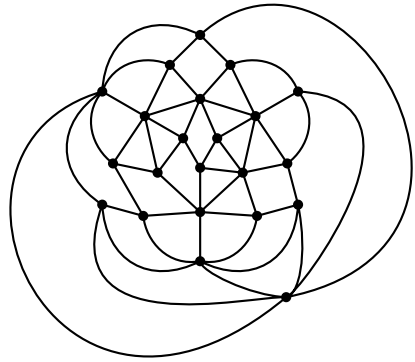


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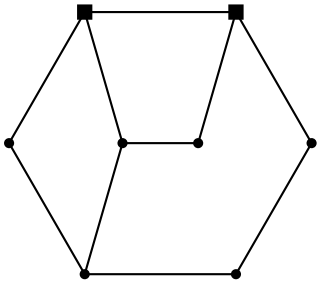


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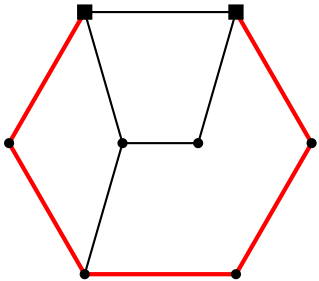
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T-joins



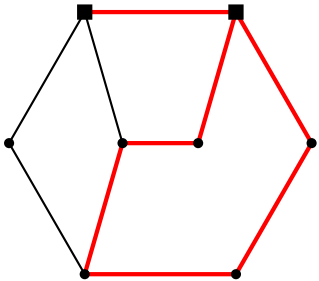
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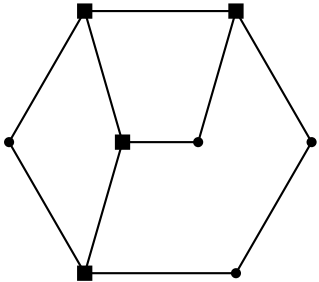
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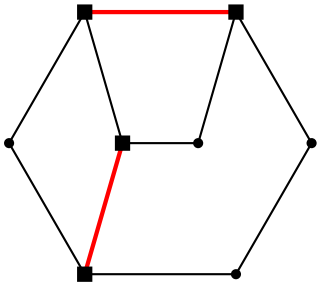
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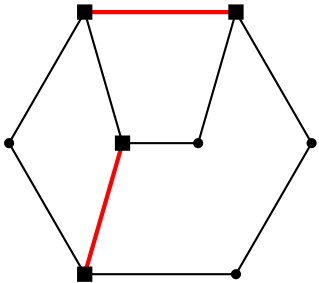
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$\tau(G,T) = 2$

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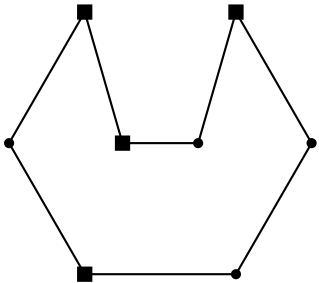
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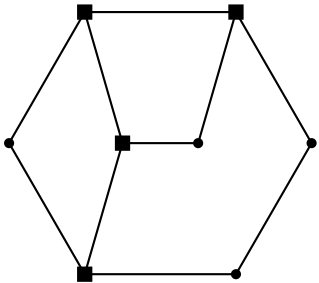
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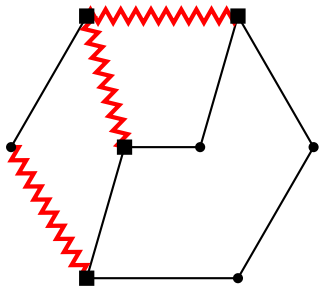
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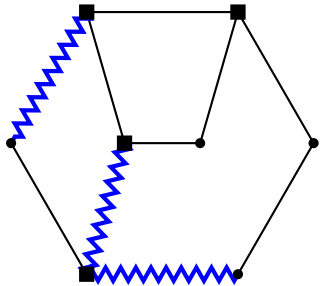
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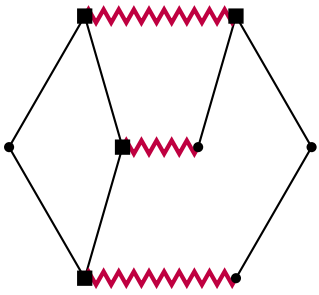
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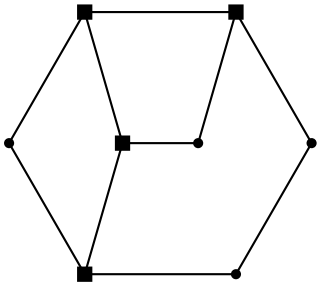
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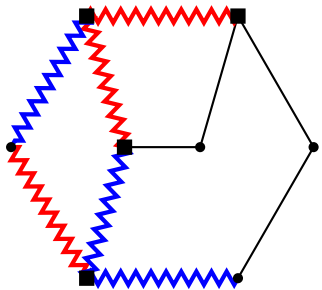
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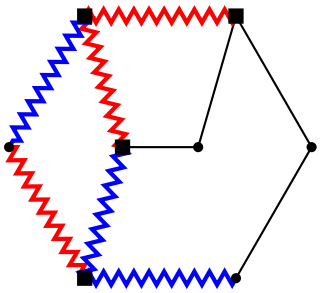
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- The maximum size of a packing of T-cuts is denoted by $\nu(G,T)$.

Relation between $\tau(G,T)$ and $\nu(G,T)$

- Every T-join intersects every T-cut, so $\tau(G,T) \geq \nu(G,T)$.

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- Every T-join intersects every T-cut, so $\tau(G,T) \geq \nu(G,T)$.
- Equality does not always hold.
- If G is bipartite, we have equality:

Theorem (Seymour 1981)

If G is bipartite and $T \subseteq V(G)$ is even, then $\tau(G,T) = \nu(G,T)$.

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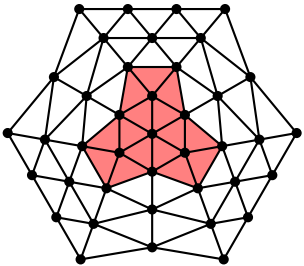
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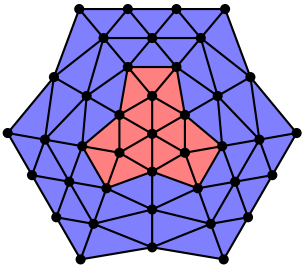
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- Use Seymour's theorem to deduce a bound on $\tau(K, T) = \tau_{\text{odd}}(G)$.

Patches and moats



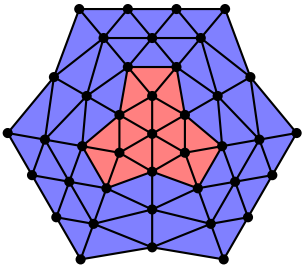
- A c-patch is an induced contractible subcomplex $L \subset K$ containing exactly c vertices of degree 5.

Patches and moats



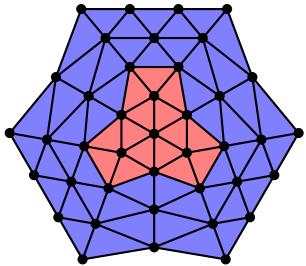
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Moat Inequality

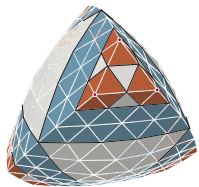
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Packing moats

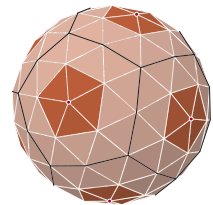
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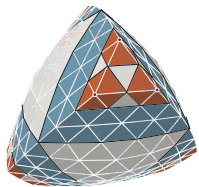
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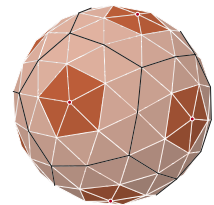
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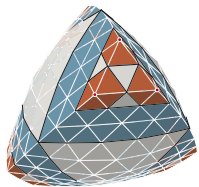
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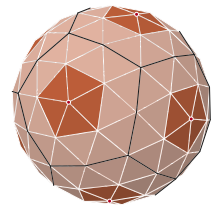
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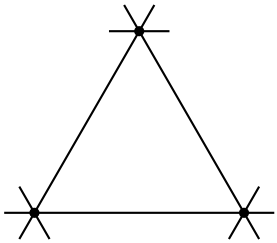
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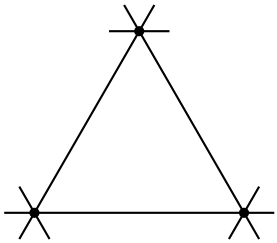
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Refinement of a plane triangulation



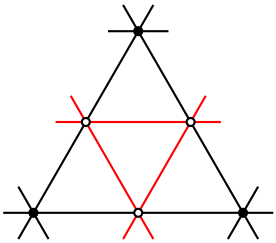
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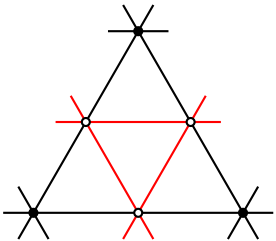
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- The *refinement* $\hat{K}$ of a plane triangulation K is constructed by subdividing each face into four faces.
- Seymour's theorem easily implies the following.

Lemma

For every planar triangulation K and every even subset $T \subseteq V(K)$,

$$\tau(K,T) = \frac{1}{2}\nu(\hat{K},T).$$

Proof of the main theorem

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- By taking the dual, we get $\tau_{\mathrm{odd}}(G) \leq \sqrt{12n/5}$.

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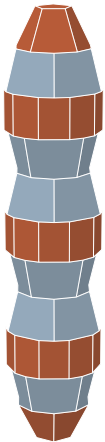
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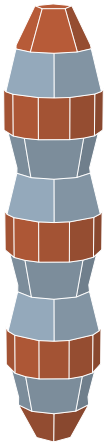
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- This shows that $\nu_{\text{odd}} = O(\sqrt{n})$ does not hold if we allow faces of size greater than 6.

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Proof of Graffiti's conjecture

- Andova et al. also showed that $\mathrm{diam}(G) \leq \frac{1}{5}n + 1$, and α and diam are integers, so

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Diameter of fullerene graphs

Conjecture (Andova and Škrekovski 2013)

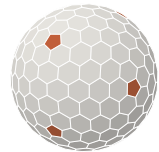
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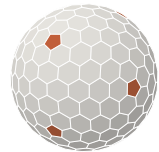
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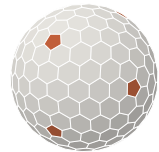
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A counterexample to the Andova–Škrekovski conjecture

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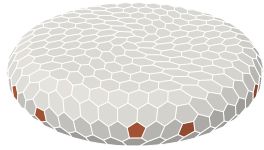
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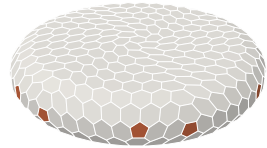


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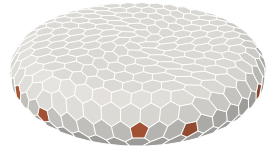
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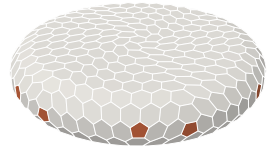
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Obrigado!